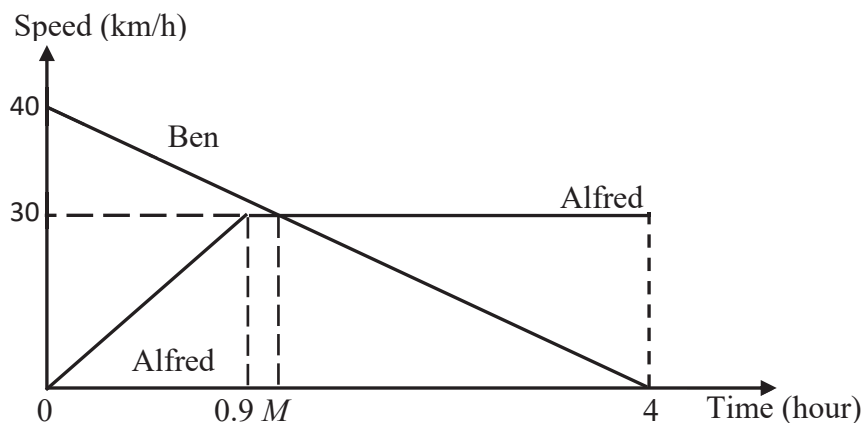


- 16 The speed-time graph shows the journeys for two drivers, Alfred and Ben. They travelled on the same road. Ben drove past Alfred at the time when Alfred started his journey.



- (a) Charles commented that Alfred will meet Ben at time = M hour since the graphs intersect at time = M hours. Is Charles correct? Explain.

[2]

- (b) Calculate the value of M .

Answer _____ h [2]

- 17 $\xi = \{\text{integers } x : 2 \leq x < 10\}$
 $E = \{\text{even numbers}\}$
 $M = \{\text{multiples of } 3\}$
 $F = \{\text{factors of } 48\}$

(a) List the element(s) in

(i) $M \cap E$

Answer _____[1]

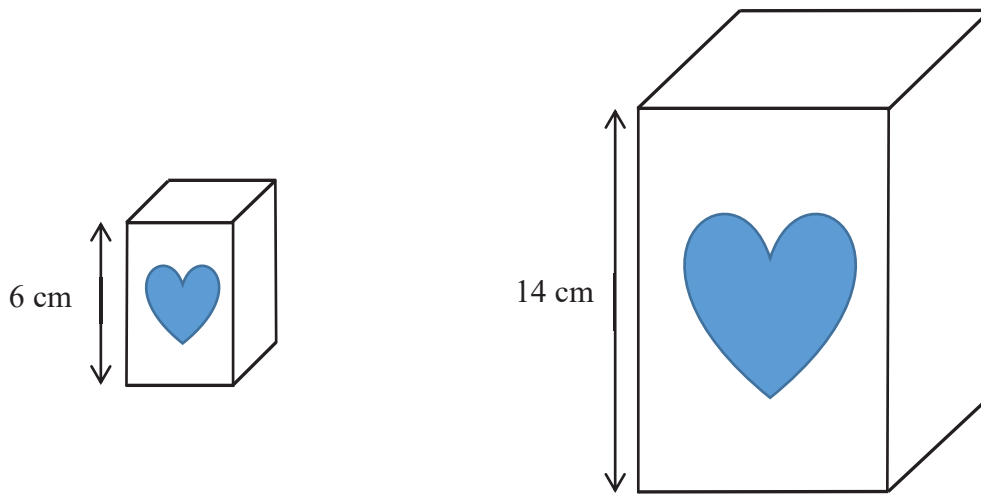
(ii) $(M \cup F)'$

Answer _____[2]

- 18 Cindy deposited \$20000 in a bank account which offered an interest of 1.2% per annum compounded monthly. She withdrew her money after 1 year 6 months. Calculate the amount of interest earned.

Answer \$ _____[2]

19. The diagram shows two geometrically similar figures made of the same material.
The length of the smaller figure is 6 cm.
The length of the larger figure is 14 cm.



- (a) The area of the heart in the larger figure is 15 cm^2 .
Calculate the area of the heart in the smaller figure.

Answer _____ cm^2 [2]

- (b) If the mass of the smaller figure is 200 grams, calculate the mass of the larger figure.

Answer _____ g [2]

- (c) The smaller figure is sold for \$2.50 while the larger figure is sold for \$ 25.50. Which figure will be a better buy? Explain.

Answer _____ [2]